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Seminar (July-December 2026)

Abstract

MCA III/ I Sem / MSc (CS) I Sem / MTech (CS/AI) I Sem/
BSc (CS) (Hons + Hons Research) / BCA (Hons + Hons Research)

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Programme:	MCA (III SEM)		
Title:	**QML Unleashed: Superposition & Entanglement**		

QML Unleashed: Superposition & Entanglement discusses concepts of superposition and entanglement, both fundamental principles of quantum computing, and their implication on Quantum Machine Learning (QML). A classical computer stores information in bits, while the quantum one stores information in quantum bits, or qubits, which are able to be in multiple states at the same time, a phenomenon called superposition. Furthermore, with the help of quantum mechanics, we can entangle two or more qubits so that the state of each qubit depends on the state of the others.

In this seminar, we explore concepts of superposition and entanglement through their implementation in a quantum circuit, as well as the unique phenomena displayed by them. Simple quantum gates such as Hadamard, Pauli-X, and CNOT gates can be used to observe superposition and entanglement, while Bell and GHZ states can be used to display the difference between classical and quantum computing and we also explore **about Data Encoding methods that use in Quantum Machine Learning.** With Qiskit, we can observe these principles through actual implementation in a quantum circuit. Finally, we discuss the applications of superposition and entanglement in QML algorithms such as quantum neural networks, variational quantum classifiers, as well as touch on the implications of quantum computing in machine learning in **Data Encoding** such as Angle Encoding , Amplitude Encoding or Basic Encoding that use in QML.

In conclusion, From this seminar we have learned that the quantum computer can process huge amounts of data in incredibly fast time periods, therefore being much more efficient than normal computers.

Keywords: Quantum Machin Learning(QML) , Superposition , Entanglement , Qubits , Data Encoding , Quantum Gates(Pauli – gates),Bell states, GHZ states, Quantum neural networks(QNN), Variational quantum classifiers(VQC).

Date:

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